
JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Counterfactual explanations (CE) are most useful when they achieve a target prediction with a small and interpretable set of input changes, but existing generators often return counterfactuals with more feature edits than are needed. We introduce *COLA*, a model- and generator-agnostic post-hoc framework that takes an existing CE, computes an optimal-transport (OT) coupling between factual and counterfactual sets, and uses the resulting alignment in a Shapley attribution rule (*p-SHAP*) to select the edits that contribute most directly to the target effect. Theoretical analysis shows that OT minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes and, under mild conditions, the refined counterfactuals do not move farther from factuals than the original CEs. Empirically, the comparison spans four datasets, twelve predictive models, and five CE generators; under this broad evaluation, *COLA* preserves the target effect using only 26–45% of the original feature edits. On the tractable small-scale benchmark, its performance is near-optimal relative to the exact mixed-integer solution.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) supports transparent and trustworthy artificial intelligence systems by making model behavior easier to inspect and compare (Arrieta et al., 2020; Das & Rad, 2020). Within this area, feature attributions (FA) methods such as Shapley values quantify how much each input feature contributes to a machine learning (ML) model output (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017); in healthcare, for example, such scores can highlight factors such as age or medical history that drive a prediction (Ter-Minassian et al., 2023; Nohara et al., 2022). Counterfactual Explanations (CE) methods answer a complementary quantitative question: how much must the input change to obtain a different model outcome (Wachter et al., 2017; Guidotti, 2022)? The large number of proposed CE algorithms (Guidotti, 2022; Verma et al., 2020) reflects substantial variation in this comparison target. Some methods generate one counterfactual per factual instance, others optimize for groups of instances, global methods summarize dataset-level directions toward a desired output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024), and distributional methods seek a counterfactual distribution that remains close in shape and cost to the factual distribution (You et al., 2025). Model assumptions also differ: some methods require differentiability, whereas others are specialized for tree-based or ensemble predictors.

Problem Description and Challenges Building on the related literature summarized in Appendix A and Appendix A’s comparison to prior CE-attribution methods, we study a unified quantitative refinement problem. FA and CE each provide useful evidence, but their weaknesses are complementary: FA ranks influential features without necessarily producing an actionable edit set, whereas a CE can reach the desired outcome while changing more features than necessary. The central question is therefore how to preserve a counterfactual effect while reducing the number and magnitude of feature modifications.

Three challenges make this comparison nontrivial. First, because CE objectives vary across instance-level, group-level, and distribution-level settings, no single generator can serve as a universal baseline. Second, a useful refinement method should compare outputs across many model classes without requiring differentiability or other special structure. Third, feature-importance scores can be quan-

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$			$\mathbf{z}'$			$\mathbf{z}''$			$\mathbf{y}^*$
$\textcircled{e}$	$\textcircled{c}$	$\textcircled{R}$	$\textcircled{e}$	$\textcircled{c}$	$\textcircled{R}$	$\textcircled{e}$	$\textcircled{c}$	$\textcircled{R}$	$\textcircled{R}$
200	5	No	250	8	Yes	200	5	No	Yes
150	3	No	150	3	No	150	7	Yes	Yes
100	2	No	350	9	Yes	100	2	No	Yes
150	6	No	150	6	No	350	6	Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] The platform compares two action plans for increasing registrations from the factual data $\mathbf{x}$, where no users are registered, toward the target outcome $\mathbf{y}^*$. Both $\mathbf{z}'$ and $\mathbf{z}''$ achieve one-half of the desired counterfactual effect by changing user-interaction features such as spending ($\textcircled{e}$) and clicks ($\textcircled{c}$), but $\mathbf{z}''$ attains the same effect with fewer feature modifications than $\mathbf{z}'$.

titatively misaligned with the edits that actually preserve counterfactual effect: performing FA independently of CE can select features that are globally important but ineffective for the specific counterfactual transition. Result II in Section 6 shows this pattern empirically, where decoupled attributions lead to weaker action plans despite modifying features identified as important.

Main Contributions We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a post-hoc framework that makes CE refinement comparable across generators and predictive models by using an optimal transport (OT)-induced coupling between factual and counterfactual sets to guide Shapley attributions. The resulting p -SHAP attribution rule includes several common Shapley variants as special cases under different couplings, which makes the role of the factual-counterfactual alignment explicit in each comparison. Theoretical results show that OT minimizes an upper bound on the W_1 outcome divergence and that, under mild conditions, the refined counterfactuals remain no farther from factuals than the original counterfactual proposals. Empirically, across four datasets, twelve models, and five CE methods, COLA with p -SHAP preserves the target effects while using only 26–45% of the original feature edits; on a small-scale benchmark, the achieved values are near the exact optimum.

To the best of our knowledge, this is the first systematic method for the problem illustrated in Figure 1 that keeps the comparison independent of any particular CE generator or model family.

2 PROBLEM FORMULATION

We now formalize the comparison in Figure 1. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ denote a black-box ML model, and let $\mathbf{x} \in \mathbb{R}^{n \times d}$ be the factual data with $n \geq 1$ instances and $d \geq 1$ features. The target model output is $\mathbf{y}^* \in \mathbb{R}^m$ with $m \geq 1$. The optimization seeks refined counterfactual data $\mathbf{z} \in \mathbb{R}^{n \times d}$ that are allowed to modify at most C feature positions while making $f(\mathbf{z})$ as close as possible to $\mathbf{y}^*$. Let D be the divergence used to quantify the gap between the compared quantities. The resulting sparse-refinement problem is:

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) and proximity constraint equation (1b) encode the standard CE trade-off: the refined counterfactual should move the model output toward $\mathbf{y}^*$ while staying close to the factual data $\mathbf{x}$, and equation (1b) makes that proximity condition explicit. This formulation leaves the metric explicit; depending on the evaluation condition, D may be Euclidean distance, OT, maximum mean discrepancy (MMD), or a mean- or median-based difference in model outcomes. The additional constraints equation (1c)–equation (1e) make the action budget measurable. The binary indicator

$\mathbf{c}$ records which row-feature positions are changed, equation (1c) limits the total number of such changes to C , and equation (1d)–equation (1e) enforce $z_{ik} = x_{ik}$ whenever $c_{ik} = 0$. When $c_{ik} = 1$, the sufficiently large constant M permits z_{ik} to take the replacement value selected by the refinement method.

The notation separates the empirical roles used throughout the comparisons: $\mathbf{x}$ is the factual reference set, $\mathbf{r}$ is the CE generator output, and $\mathbf{z}$ is the refined action plan evaluated under the same metric. Thus $\mathbf{x}$ provides the baseline for measuring proximity, $\mathbf{r}$ provides candidate counterfactual values, and $\mathbf{z}$ records which candidate edits remain after the action budget is applied. The effect comparison uses $f(\mathbf{x})$, $f(\mathbf{r})$, and $f(\mathbf{z})$ against $\mathbf{y}^*$, while the distance comparison reports how far $\mathbf{z}$ moves from $\mathbf{x}$ relative to the movement from $\mathbf{x}$ to $\mathbf{r}$. When the coupling $\mathbf{p}$ is used, it aligns $\mathbf{x}$ and $\mathbf{r}$ before values are assigned to $\mathbf{z}$; the resulting $\mathbf{z}$ is then compared with both $\mathbf{x}$ and $\mathbf{r}$ under the same evaluation condition. In tables and curves, each metric keeps the factual side fixed as $\mathbf{x}$: proximity is measured from $\mathbf{x}$, normalization uses the movement out of $\mathbf{x}$, and target recovery asks whether edits selected from $\mathbf{x}$ produce a stronger refined outcome. We therefore use $\mathbf{x}$ consistently for the factual baseline, while $\mathbf{z}$ denotes the post-refinement candidate and $\mathbf{z}$ ’s achieved counterfactual effect.

Because equation (1) directly compares counterfactual effect against an action budget, we use FA to score the row-feature positions that should receive the limited modifications encoded by $\mathbf{c}$. The next section reviews Shapley-value baselines that are later compared with our proposed attribution rule inside COLA to construct the refined counterfactual $\mathbf{z}$ from factual inputs $\mathbf{x}$, counterfactual references $\mathbf{r}$, and the target output $\mathbf{y}^*$. The sparse optimization is computationally difficult even for linear models with $d = 1$; see Appendix B, and Algorithm 1 gives the scalable heuristic used in the experiments.

3 PRELIMINARIES ON SHAPLEY VALUE

This section establishes the Shapley baselines that will later be compared inside COLA. We first recall the standard game-theoretic definition and then distinguish the baseline, random-baseline, and counterfactual versions by the reference distribution each method uses for attribution. A broader survey of Shapley-value attribution methods is provided by (Sundararajan & Najmi, 2020).

In cooperative game theory, a coalitional game is specified by a finite player set—features in our setting— $\mathcal{F} = \{1, 2, \dots, d\}$ and a characteristic function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$. For each subset $S \subseteq \mathcal{F}$, the value $v(S)$ measures the payoff obtained when only the features in S are present, with $v(\emptyset) = 0$. The Shapley value uses these subset-level comparisons to allocate the total payoff $v(\mathcal{F})$ across individual features.

For a feature k and a coalition S , let $\Delta(k, S)$ denote the marginal contribution of adding k to S . This quantity is the payoff difference between the coalition with and without feature k :

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

Averaging this marginal contribution over all subset sizes gives the Shapley value for feature k , defined in equation (3).

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

Thus, ϕ_k is a normalized average of feature k ’s marginal effect across every subset S that excludes k . The inverse binomial term weights coalitions of the same size equally, and the final division by d yields an axiomatic allocation of the total payoff across features.

For a data point $\mathbf{x}_i$, $\mathbf{x}_{i,S}$ denotes the coordinates in S , while $\mathbf{x}_{i,\mathcal{F} \setminus S}$ denotes the remaining coordinates; we use the same convention for other vectors. The following variants differ in the baseline or counterfactual reference used to evaluate each subset comparison.

Baseline Shapley (B-SHAP). This method compares each factual point $\mathbf{x}_i$ to a specific reference point $\mathbf{r}_j$, so the attribution depends on an explicit one-to-one baseline choice. The set function is

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j,\mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

Here, a missing feature is replaced by the corresponding coordinate of the reference point $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). The comparison therefore assumes a known exact alignment between $\mathbf{x}$ and $\mathbf{r}$.

Random Baseline Shapley (RB-SHAP). RB-SHAP replaces the fixed reference with an expectation over a background distribution, a choice used implicitly in (Lundberg & Lee, 2017) and the widely used “SHAP” library. Its set function is

$$v_{\text{RB}}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ is the background distribution, commonly instantiated as the training data (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). CF-SHAP makes the baseline contrast explicitly counterfactual by setting the reference distribution to the counterfactual distribution associated with each factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$).

$$v_{\text{CF}}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This method has been adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021) and is empirically useful for contrastive FA. Its quantitative behavior depends on the probabilistic alignment assumed between factials $\mathbf{x}$ and counterfactuals $\mathbf{r}$.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We generalize equation (4)–equation (6) by making the factual-counterfactual coupling an explicit input. The algorithm A_{Prob} returns the joint probability used to compare factual rows with counterfactual rows.¹ With this coupling fixed, p -SHAP is defined as follows,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

The key quantitative distinction is that p -SHAP evaluates attribution under a chosen joint distribution rather than under an unstructured random baseline. When the joint distribution is obtained by OT, the attribution is tied to the least-cost factual-counterfactual coupling used in the refinement comparison.

Importantly, A_{Prob} is independent of the CE algorithm, but only depends on the generated counterfactual $\mathbf{r}$. This separation keeps the attribution comparison fixed after $\mathbf{r}$ is generated, rather than inheriting variability from the CE generation mechanism itself. As Result II in Section 6 shows, the OT coupling can outperform a counterfactual distribution defined by the CE generation process as in Albini et al. (2022), highlighting that the alignment used for attribution is an empirical design choice with measurable consequences.

A central instantiation in this paper uses the following OT problem, equivalently the entropic form of the 2-Wasserstein comparison. The resulting transport plan $\mathbf{p}_{\text{OT}}$ is then used as the joint distribution between $\mathbf{x}$ and $\mathbf{r}$ in p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Here, $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ are the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of all admissible joint distributions, or transport plans. The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is entropic regularization with coefficient $\varepsilon \geq 0$; when $\varepsilon > 0$, it improves computational efficiency while preserving the same alignment role in the attribution pipeline.

¹First, p -SHAP degrades to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates an $i \leftrightarrow j$ alignment of for any $\mathbf{x}_i, \mathbf{r}_j$. Second, p -SHAP degrades to RB-SHAP in equation (5) when A_{Prob} is defined to be independent of CE but associates with an arbitrary distribution $\mathcal{D}$. Third, p -SHAP degrades to CF-SHAP in equation (6) when A_{Prob} is built upon a known distribution of CE.

Theoretical Aspects of p -SHAP OT directly optimizes the total feature-modification cost for moving factual data $\mathbf{x}$ toward counterfactual data $\mathbf{r}$ under the selected alignment. This matches the empirical criterion used throughout the paper: achieving the counterfactual outcome with fewer and smaller edits. Theorem 4.1 (proof in Appendix C) makes this connection quantitative under Lipschitz continuity of the predictive model f . It shows that the OT plan $\mathbf{p}_{\text{OT}}$ used in p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$. Specifically, the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant times the square root of the expected feature-modification cost, $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, where p_{ij} quantifies how strongly $\mathbf{x}_i$ is aligned to one or more counterfactual rows $\mathbf{r}_j$. Operationally, $\mathbf{p}_{\text{OT}}$ supplies the lowest-cost alignment among all feasible plans ($\mathbf{p} \in \Pi$). As a result, the same coupling that drives p -SHAP also gives the tightest upper bound, within this analysis, on the remaining counterfactual-effect gap $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Theorem 4.1 (p -SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{\text{OT}}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Although Theorem 4.1 controls transport cost rather than the discrete L_0 action count directly, it provides a convex proxy for the $\mathcal{NP}$ -hard sparsity objective. By concentrating attribution mass on lower-cost factual-counterfactual paths, the OT coupling gives Algorithm 1 a principled basis for discarding lower-yield feature edits. Theorem 4.2 below (proof in Appendix D) further clarifies the interpretation of p -SHAP: the subset score corresponds to an interventional contrast in the expected counterfactual model output.

Theorem 4.2 (Interventional Effect of p -SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Finally, p -SHAP preserves the same Shapley axioms used by B-SHAP and RB-SHAP, so the improved empirical comparisons later in the paper come from the coupling choice rather than from abandoning the standard attribution framework. We omit the proof and refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) for the Shapley axioms.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch Algorithm 1 summarizes COLA as a four-stage refinement pipeline for equation (1): generate an initial CE, compute a factual-counterfactual coupling, score row-feature edits, and replace only the highest-scoring positions. The line-by-line description below makes explicit which component controls each comparison in the experiments; Figure 2 illustrates the same flow.

Line 1 (Applying a CE algorithm to find a counterfactual $\mathbf{r}$). The CE generator A_{CE} receives the model f , factual data $\mathbf{x}$, target output $\mathbf{y}^*$, and tolerance ϵ . It returns the reference counterfactual $\mathbf{r}$ that subsequent methods all try to match or approximate, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.

Line 2 (Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$). The coupling algorithm A_{Prob} takes $\mathbf{x}$ and $\mathbf{r}$ and outputs $\mathbf{p} \in \mathbb{R}_+^{n \times m}$, a joint distribution over factual and counterfactual rows. This matrix determines which counterfactual rows are used as the quantitative references for each factual row in Line 5. As reviewed in Section 3, when the true alignment is unknown, the OT choice computes the coupling with the smallest transport cost between $\mathbf{x}$ and $\mathbf{r}$; when a reliable alignment is known, it can be supplied directly and compared as a separate condition.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

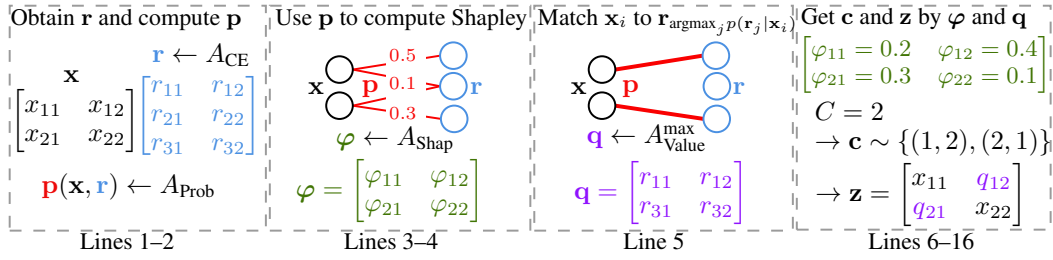


Figure 2: [An illustration of COLA] The pipeline converts an initial counterfactual set into an action indicator $\mathbf{c}$ and refined counterfactual $\mathbf{z}$ for equation (1). For clarity, the example uses $A_{\text{Value}}^{\max}$ in line 5. With budget $C = 2$, lines 6–16 select exactly two row-feature positions according to the normalized attribution matrix φ and replace only those values.

Lines 3–4 (p -SHAP FA). We use equation (7) to compute Shapley values for the factual matrix $\mathbf{x}$. The coupling $\mathbf{p}$ determines which rows of $\mathbf{r}$ act as the contrastive references for each factual row, making the alignment an explicit experimental factor rather than an implicit implementation detail. After computing the Shapley matrix, COLA takes element-wise absolute values and normalizes them to sum to one. The resulting matrix φ is therefore a probability distribution over candidate row-feature edits.

Line 5 (Computing feature values). The value-assignment algorithm A_{Value} uses $\mathbf{r}$ and $\mathbf{p}$ to build the replacement matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$. For each factual row, it selects either one counterfactual reference row or an average over multiple reference rows, so each q_{ik} is the candidate value used if feature k of row i is selected for modification. The two instantiations below separate these single-reference and weighted-reference conditions.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\max}$ chooses the counterfactual row with the largest coupling probability.

$$\mathbf{q} = A_{\text{Value}}^{\max}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i), k} \text{ and } \tau(i) = \arg \max_{j=1, 2, \dots, m} p_{ij}. \quad (9)$$

The algorithm $A_{\text{Value}}^{\text{avg}}$ instead computes q_{ik} as a coupling-weighted convex combination of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Because φ is nonnegative and sums to one, COLA treats it as the edit-selection policy under the budget C . Line 7 samples the modified positions by setting $c_{ik} = 1$, and each selected factual value x_{ik} is replaced by its candidate q_{ik} . The output pair $(\mathbf{c}, \mathbf{z})$ therefore reports both the number and location of edits and the resulting refined counterfactual used in equation (1).

By Theorem 4.1, COLA targets the empirical quantity $D(f(\mathbf{z}), \mathbf{y}^*)$ by assigning the limited edits to features with the largest attribution-supported contribution to the counterfactual effect. The next theorem (proof in Appendix E) provides the corresponding proximity guarantee: the refined $\mathbf{z}$ produced by COLA is no farther from the factual data than the aligned counterfactual proposal under the Frobenius norm. Table 3 reports the same practical pattern through the normalized feature-change ratios.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the facts as the original counterfactual set reordered by σ .

Complexity of COLA The runtime separates into the same components used in the empirical comparisons. Let $\mathcal{O}(M_{\text{CE}})$ be the cost of the chosen CE generator A_{CE} . When A_{Shap} estimates Shapley values by weighted linear regression with M_{Shap} sampled subsets, the overall complexity as a function of n, m, d , and the entropic-OT regularization parameter ε is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where the value-assignment term is $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ for $A_{\text{value}}^{\text{max}}$ and $N = \mathcal{O}(nmd)$ for $A_{\text{value}}^{\text{avg}}$. Appendix F gives the detailed derivation.

6 NUMERICAL RESULTS

This section evaluates COLA on equation (1) by comparing the refined counterfactual $\mathbf{z}$ against the CE-generator output $\mathbf{r}$, with $\mathbf{y}^* = f(\mathbf{r})$. We report results under four divergence metrics so that the empirical patterns are not tied to a single notion of discrepancy: OT compares full distributions, MMD compares distributions through high-dimensional mean embeddings, and absolute mean difference (MeanD) and absolute median difference (MedianD) compare outcome means and medians, respectively. The numerical evidence is organized around three claims: I) COLA substantially reduces the number and magnitude of feature modifications. II) p -SHAP outperforms the Shapley baselines under matched evaluation conditions. III) COLA is near-optimal on tractable benchmarks with an exact solver.

Experiment Setup The benchmark covers 4 binary-classification datasets, 5 CE generators with different design goals, and 12 predictive models. As summarized in Table 1, each dataset– A_{CE} –model triple defines one experiment scenario, which lets us compare methods under matched data, generator, model, metric, and action-budget conditions.

The CE generators in Table 1 span instance-level, group-level, and distribution-level baselines, and the setup in Table 1 fixes the dataset–model–generator combinations used for the quantitative comparisons. DiCE and KNN generate counterfactuals for individual factual instances; AReS and GlobeCE produce a collection of counterfactual instances for the factual group; and Discount treats the factual sample as an empirical distribution and searches for a nearby counterfactual distribution.

Table 2 defines the six attribution-and-coupling variants compared throughout the experiments. CF- p_{OT} is the proposed p -SHAP instantiation, while the other rows are baselines that isolate the effect of

Table 1: [Experiment Scenarios Setup] Scenario grid for the empirical comparisons. The benchmark combines four datasets, five CE algorithms, and 12 models; one “scenario” is a fixed dataset, A_{CE} algorithm, and model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] Six compared methods, with colors matched to Figure 3. Each row fixes the coupling $\mathbf{p}$ used by A_{value} and/or A_{Shap} , and each method is evaluated within the scenarios defined in Table 1.

Method	The probability $\mathbf{p}$ used by A_{value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

the background distribution, counterfactual information, and alignment. Each method is specified by A_{Prob} , A_{value} , and A_{Shap} ; for example, RB- p_{Uni} uses a uniform coupling for $\mathbf{p}$, passes that coupling to equation (9) or equation (10) to compute $\mathbf{q}$, and uses RB-SHAP for attribution.

The six methods form targeted ablations. RB- p_{Uni} versus CF- p_{Uni} isolates the value of using counterfactual rather than training-set information. CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} hold counterfactual attribution fixed while changing only the joint distribution, allowing the OT alignment to be compared against uniform and random alignments. RB- p_{OT} versus CF- p_{OT} tests whether OT helps because it informs A_{Shap} , not merely because it is used downstream by A_{value} . Finally, CF- p_{Ect} uses the known source of each generated counterfactual as an exact factual-counterfactual alignment, making the comparison coincide with CF-SHAP or B-SHAP in that special case.

Result I: COLA achieves significant action reduction with a minor loss in counterfactual effect

Table 3 compares methods by the edit budget needed for $f(\mathbf{z})$ to recover either 80% or 100% of the counterfactual effect achieved by $f(\mathbf{r})^2$. Across datasets, the refined counterfactuals require substantially fewer modified features than the original CE $\mathbf{r}$ while still reaching the specified effect threshold. At the 80% threshold, COLA uses only 13%–25% of the original CE movement as measured by $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$. At the stricter 100% threshold, only COLA with p -SHAP (CF- p_{OT}) consistently reaches the target, doing so with 26%–45% of the original feature-change magnitude. These margins make CF- p_{OT} the strongest method for action minimality in the table.

Result II: p -SHAP outperforms the other Shapley methods in achieving counterfactual effect

Figure 3 compares the Shapley variants in equation (4)–equation (7) under the same action budgets. The x-axis is the allowed number of feature changes C , and the y-axis is the remaining divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ from equation (1); lower values at the same C therefore indicate better use of the edit budget. The RB baselines, RB- p_{Uni} and RB- p_{OT} , remain consistently worse, showing that training-set background information alone is insufficient for sparse CE refinement. CF- p_{OT} also improves over RB- p_{OT} , and because these methods differ only in A_{Shap} (Table 2), the margin isolates the benefit

²By definition, $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, so $\mathbf{r}$ achieves 100% counterfactual effect because $\mathbf{y}^* = f(\mathbf{r})$. For the 80% condition, we measure the proportion of divergence reduced by the refined CE $\mathbf{z}$: *Counterfactual Effect* = $1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D being OT.

Table 3: [COLA for Modification Minimality] Action counts and normalized movement for each method. Entries report the number of modified features in $\mathbf{z}$ required to reach either 80% or 100% of the counterfactual effect of $f(\mathbf{r})$, averaged over 4 randomly selected scenarios from Table 1 with 100 runs per scenario. The ratio $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$ compares the magnitude of the refined edits to the original CE edits; “-” indicates that the target effect was not reached.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z} - \mathbf{x}\ }{\ \mathbf{r} - \mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z} - \mathbf{x}\ }{\ \mathbf{r} - \mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

of using the OT coupling inside attribution rather than only in value assignment. Finally, CF- p_{OT} improves over CF- p_{Uni} and CF- p_{Rnd} , demonstrating that counterfactual information is most effective when paired with a structured factual-counterfactual alignment rather than a uniform or random joint distribution.

The same ordering appears across many scenarios in Table 1, both in the action-count summary of Table 3 and in the full budget curves of Figure 3. Appendix H provides additional curves, which support the same empirical pattern: among the compared Shapley-based refinements, p -SHAP gives the most favorable sparsity–effect trade-off for CE refinement.

Result III: COLA is competitive with the mixed integer linear programming (MILP) optimum on tractable benchmarks Figure 4 compares COLA variants against an exact MILP solution on the subset of settings where that benchmark is computationally tractable, and Figure 4 is therefore the direct optimality check rather than a scalability benchmark. This result tests whether p -SHAP can reduce dependence on CE-generator-specific alignment knowledge³ by replacing it with the OT joint distribution between factual and counterfactual data. The comparison focuses on equation (1) with MeanD as the divergence function D . CF- p_{Ect} uses the known factual-counterfactual alignment, while the exact optimum is obtained from the MILP formulation in Appendix G. Because solving the MILP is computationally expensive, this optimality comparison is restricted to *German Credit*.

The DiCE panels show that CF- p_{Ect} is close to the optimum, indicating that COLA can exploit exact instance-level alignments when they are available. Since DiCE and KNN produce independent factual-counterfactual pairs, these results also show that the general formulation in equation (1) remains effective in instance-based CE settings, not only in group or distributional settings. Notably, CF- p_{OT} sometimes improves over CF- p_{Ect} , suggesting that the OT coupling can provide a stronger attribution alignment than CE-generator-dependent pairing information. In the KNN cases, however,

³This knowledge, i.e. exact alignment between factual and counterfactual, is available only by DiCE and KNN among all CE methods considered.

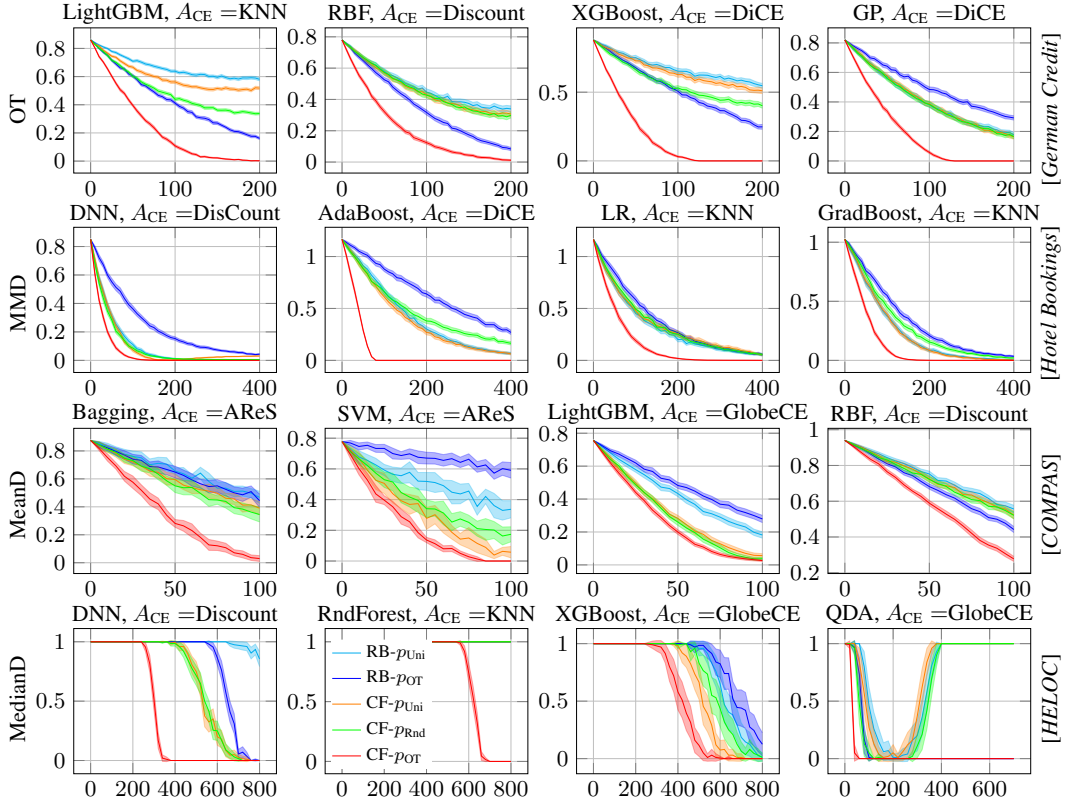


Figure 3: Counterfactual-effect gap $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the allowed action budget C . Each curve summarizes 100 runs, with shadows indicating 99.9% confidence intervals; lower curves indicate that a method reaches the target effect with fewer allowed edits. $A_{\text{Value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, and $A_{\text{Value}}^{\text{max}}$ is used for *German Credit* and *Hotel Bookings*. The legend inside “RndForest, $A_{\text{CE}} = \text{KNN}$ ” applies to all plots.

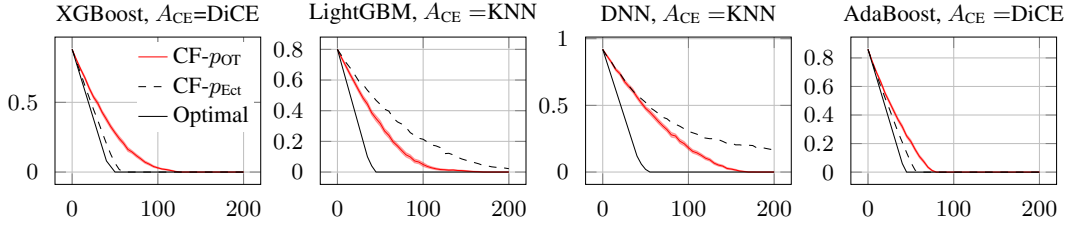


Figure 4: [*German Credit*] Comparison with the tractable optimum: $D(f(\mathbf{z}), \mathbf{y}^*)$ versus the allowed action budget C , using *MeanD* as D . Lower values indicate closer recovery of the target outcome for the same number of edits.

both $\text{CF-}p_{\text{Ect}}$ and $\text{CF-}p_{\text{OT}}$ remain separated from the exact optimum, so identifying the best alignment remains an open empirical and theoretical question.

7 CONCLUSIONS

This paper introduced COLA, a post-hoc framework for refining CE with joint-distribution-informed Shapley values. Across the empirical comparisons, the proposed $\text{CF-}p_{\text{OT}}$ instantiation preserves counterfactual effect with substantially fewer edits than the original CE and the Shapley baselines, using only 26–45% of the original feature-change magnitude when the full target effect is recovered. The theoretical results explain why the OT coupling is a favorable alignment for this comparison: it

minimizes a cost-based upper bound on outcome divergence and keeps the refined counterfactual close to the factual input. Together, the results show that making the factual-counterfactual alignment explicit improves both the clarity of the attribution and the practical sparsity of the resulting action plan.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper proposes *COLA* with *p*-SHAP, a methodological and empirical framework for reducing the number of edits in CE while preserving measured counterfactual effect. The study does not involve human subjects or interventions, and we did not collect, annotate, or release any new personal data.

All empirical comparisons use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We follow dataset licenses/terms and use these data solely for benchmarking method behavior across datasets, models, CE generators, and metrics. We do not attempt re-identification or linkage, and we release no additional personal information. Our code will include options to exclude protected attributes and to enforce feasibility/immutability constraints on features.

CE can surface biases in underlying models but do not by themselves guarantee fairness. Although our method reduces the measured number of recommended feature edits, it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age), nor does it establish causal validity. We discourage using CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavior changes. Any deployment in consequential domains should include domain-expert-defined feasibility constraints, fairness audits (e.g., disparate impact/benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. Our attribution is distributional rather than causal and must not be interpreted as causal effect.

The main risks are (i) misinterpreting attribution scores as causal estimates, (ii) treating sparse edits as feasible or beneficial without domain review, and (iii) enabling strategic manipulation. We mitigate these risks by stating the non-causal scope, supporting cost and feasibility masks together with immutable-feature constraints, reporting subgroup performance where relevant, and documenting responsible-use guidance.

To support empirical verification, we provide an anonymized implementation, detailed settings, and seeds; we also report model types, hyperparameters, compute budgets, and fixed evaluation protocols for all comparisons. To our knowledge, there are no conflicts of interest or sensitive sponsorships related to this submission; any funding disclosures will be provided after the double-blind review.

REPRODUCIBILITY STATEMENT

We provide the components needed to reproduce both the method and the quantitative comparisons. The COLA framework and *p*-SHAP attribution are specified with notation and pseudocode in the main paper (Algorithm 1; Method and Preliminaries sections), while the theoretical assumptions and complete proofs are given in the appendix.

Appendix I documents hyperparameters, optimization settings, environment specifications, random seeds, datasets, preprocessing, and data splits; Appendix I also reports hardware and runtime for the quantitative comparisons. The experimental sections enumerate the compared models, CE generators, coupling choices, and ablations, and the appendix includes the small-scale MILP benchmark setup used for the optimality comparison. The experiment code and software will be released upon acceptance; hardware and runtime are reported to contextualize the compute required to reproduce the results.

⁴<https://iclr.cc/public/CodeOfEthics>

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APPENDICES

A	Comparison to Existing Approaches	15
B	$\mathcal{NP}$-hardness of the Problem in equation (1)	15
C	Proof of Theorem 4.1: p-SHAP Towards Counterfactual Effect	16
D	Proof of Theorem 4.2: Interventional Effect	18
E	Proof of Theorem 5.1: Counterfactual Proximity	18
F	Analysis of Computational Complexity of COLA	19
G	An MILP Formulation of equation (1) With MeanD	20
H	Extended Numerical Results	21
I	Experiments Reproducibility	22

A COMPARISON TO EXISTING APPROACHES

CF-SHAP (Albini et al., 2022) uses counterfactual data points as the Shapley background distribution, but its comparison requires a counterfactual distribution conditioned on each factual instance. This assumption amounts to knowing a probabilistic alignment between every factual and counterfactual instance, and similar alignment assumptions appear in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). Such assumptions are restrictive for global, group, and distributional explanation settings, whereas our formulation treats the setup in (Albini et al., 2022) as one special coupling choice. WeightedSHAP (Kwon & Zou, 2022) weights feature contributions; our method instead weights contributions across data points and can be extended to weight both rows and columns. The framework in (Kommiya Mothilal et al., 2021) also combines FA and CE, but inherits the same conditional-counterfactual assumption as (Albini et al., 2022), limiting direct comparison with group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025). The method of (Sharma et al.) decomposes a single observed system-level change using causal structure and time-series predictors. None of these works, however, directly benchmark the minimal-action CE refinement problem studied here.

The action-cost formulations in (Kanamori et al., 2022; Karimi et al., 2021) are closest to our objective. (Karimi et al., 2021) minimize intervention cost under a known structural causal model (SCM), an assumption that is rarely available in practice. (Kanamori et al., 2022) emphasize that it remains open whether existing CE generators can solve this type of action problem, and propose MILP-based exact approaches for linear classifiers, tree ensembles, and deep ReLU networks, building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). Those exact baselines are valuable for optimality comparisons, but their computational cost limits scalability; this motivates our post-hoc refinement strategy and the small-scale MILP benchmark reported in the main results.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 explains why the empirical comparisons focus on scalable refinement rather than an exact solver for every scenario. Unless $\mathcal{P} = \mathcal{NP}$, the sparse-action objective is not expected to admit a generally scalable exact algorithm; the difficulty comes from the combinatorial edit budget even when the predictive model is simple.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. We reduce from *Sparse Regression (SR) with a Cardinality Constraint*. Given $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, this problem asks for a vector $\mathbf{z} \in \mathbb{R}^n$ with at most K nonzero entries that minimizes the residual error:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ counts the nonzero entries of $\mathbf{z}$. The problem is $\mathcal{NP}$ -hard because the optimizer must choose which subset of variables receives the limited nonzero budget.

We map this sparse-regression instance into Problem equation (1) as follows. Set the number of features to $d = 1$, use n instances to match the sparse-regression dimension, take the factual data to be $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$, and keep the same target output $\mathbf{y}^* \in \mathbb{R}^m$. The predictive model is linear, $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$. This construction is another instance of Problem equation (1), with $\mathbf{x}$ fixed at zero and the feasible counterfactual values represented by $\mathbf{r}$ whenever sparse coordinates are allowed to move.

Use squared Euclidean distance for the output comparison, $D(f(\mathbf{z}), \mathbf{y}^*) = \|f(\mathbf{z}) - \mathbf{y}^*\|_2^2$, and for the instance comparison, $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$ because $\mathbf{x} = \mathbf{0}$. Set the action budget C equal to the sparse-regression budget k , and choose M large enough to avoid binding changed coordinates, for example $M \geq \max_i |z_i|$.

Given that $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) are simplified to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

Thus, when $c_i = 0$, the constraints force $z_i = 0$; when $c_i = 1$, the coordinate may vary within $[-M, M]$. The action constraint equation (1c) becomes the same cardinality budget, $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thus becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \quad (11a)$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \quad (11b)$$

$$\sum_{k=1}^d c_i \leq k, \quad (11c)$$

$$z_i = 0, \quad \text{if } c_i = 0, \quad (11d)$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \quad (11e)$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \quad (11f)$$

In this reduced instance, each indicator c_i records whether coordinate z_i is allowed to move away from the factual value zero. The constraints therefore reproduce the sparse-regression cardinality condition, and the objective is exactly the sparse-regression residual.

Because this construction preserves both the residual objective and the cardinality budget, an exact solver for Problem equation (1) would solve the $\mathcal{NP}$ -hard sparse-regression problem. Hence Problem equation (1) remains $\mathcal{NP}$ -hard even with $d = 1$, a linear model f , and standard Euclidean divergences; the same comparison applies when the candidate reference $\mathbf{r}$ is viewed as the sparse vector selected by the reduction. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the factual set with weights $\mu_i \geq 0$ and $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the counterfactual set with weights $\nu_j \geq 0$ and $\sum_{j=1}^m \nu_j = 1$. The OT plan $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ is the feasible coupling that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions with marginals $\sum_{j=1}^m p_{ij} = \mu_i$ for every i and $\sum_{i=1}^n p_{ij} = \nu_j$ for every j . The theorem shows that the same coupling used for attribution also minimizes the cost term that upper-bounds the counterfactual-effect gap.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is*

Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. By Lipschitz continuity of f with constant L , every factual-counterfactual pair satisfies:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Weighting this pairwise comparison by the nonnegative OT mass p_{ij}^{OT} gives:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing over all factual and counterfactual indices yields the corresponding expected-output bound:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Denote the left-hand expected output difference by $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and the corresponding expected feature distance by $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. Then:

$$E_f \leq L E_d.$$

To express this in the squared-cost form used by OT, apply Cauchy–Schwarz. The OT weights are nonnegative and sum to one because p_{OT} is a joint probability distribution over $\mathbf{x}$ and $\mathbf{r}$. For any arrays a_{ij} and b_{ij} , Cauchy–Schwarz gives

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy–Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f^5 , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability of $\mathbf{x}$ and $\mathbf{r}$ obtained from A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are set to $\mathbf{x}_{i,S}$, and the features in $\mathcal{F} \setminus S$ remain distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Therefore, the expected value of $f(\mathbf{r})$ under the intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

Remark that by the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Hence the conclusion. $\square$

The value function $v^{(i)}(S)$ captures the expected value of $f(\mathbf{r})$ when we intervene by setting the features in S to $\mathbf{x}_{i,S}$, denoted as $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This intervention is independent of any predefined joint probability distribution $p(\mathbf{x}, \mathbf{r})$. Therefore, the expression $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ represents the combined effect of the base expected value of $f(\mathbf{r})$ and the additional causal impact of the attribution $v^{(i)}(S)$.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$. For the elements where $c_{ik} = 1$ (modified elements), $z_{ik} = q_{ik}$. For elements where $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore, we can write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Hence the conclusion. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the facts as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ becomes a permutation matrix associated with a bijection σ . Consequently, the value assignment algorithm A_{Value} (whether maximizing or averaging) assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting $\mathbf{q}$ with the permuted matrix $\mathbf{r}_\sigma$ into the general bound established in Theorem 5.1 (i.e., $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) directly yields the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

This section decomposes the computational cost of COLA according to the same modules compared in the experiments.

First consider A_{Prob} , the module that determines the factual-counterfactual coupling. If the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, constructing $\mathbf{p}$ from that alignment costs $\mathcal{O}(n \times m)$. If the alignment is estimated by entropic OT, the Sinkhorn–Knopp algorithm yields $\mathbf{p}$ in $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Next consider A_{Shap} . For each sampled subset, the model is evaluated across all n factual rows, and reference values from $\mathbf{r}$ are inserted for the features outside the subset. With precomputed reference access, the feature-replacement cost scales with d per subset. For M_{Shap} sampled subsets, this gives $\mathcal{O}(n \times d \times M_{\text{Shap}})$ for the attribution module.

The normalization of the attribution matrix in line 4 scans all row-feature entries and costs $\mathcal{O}(n \times d)$.

To compare the complexities of the two algorithms, $A_{\text{Value}}^{\max}$ and $A_{\text{Value}}^{\text{avg}}$, we analyze each algorithm step-by-step. For $A_{\text{Value}}^{\max}$, for each row $\mathbf{x}_i$ in the data table, we need to (1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, (2) identify the row $\mathbf{r}_j$ in the reference data with the highest probability p_{ij} , which involves $\mathcal{O}(m)$ operations per row, and (3) assign $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, which involves $\mathcal{O}(d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\max}$ is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, for each row $\mathbf{x}_i$ in the data table, we need to 1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, 2) compute the sum $\sum_{j'=1}^m p_{ij'}$, which involves $\mathcal{O}(m)$ operations per row, and 3) calculate the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, which involves $\mathcal{O}(m \times d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{avg}}$ is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, selecting and applying the edits costs $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$ because the budget satisfies $C \leq n \times d$.

Therefore, the complexity of COLA using $A_{\text{Value}}^{\max}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA using $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Combining these components, the complexity of COLA as a function of n , m , d , and the entropic-OT regularization parameter ε is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\max}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

This section derives the exact optimality benchmark used to quantify how close COLA is to the best sparse action plan when the factual-counterfactual alignment is known and D is MeanD:

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Since COLA is used, we have $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$, and $\mathbf{z}$ stays closer to $\mathbf{x}$ than $\mathbf{r}$, hence equation (1b) is dropped. The formulation of equation (1) then becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \tag{13a}$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \tag{13b}$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \tag{13c}$$

Note that the original constraints equation (1d) and equation (1e) merge to be equation (13c), because CF- p_{Ect} is imposed to be used. That is, for any $\mathbf{x}_i$, there is an exact $\mathbf{r}_j$ serves as its reference in A_{Shap} and A_{Value} , such that x_{ik} ($k = 1, 2, \dots, d$) either stays unchanged or can be changed to r_{jk} . Therefore $z_{ik} = r_{jk}c_{ik} + x_{ik}(1 - c_{ik})$ of which the value depends on the binary variable c_{ik} .

Due to the known alignment between any $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$, $\mathbf{q}$ is determined (also, both $A_{\text{Values}}^{\text{max}}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$). For any data point i and any feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denotes the solution $\mathbf{z}_i$ where we have all features $k \in S$ changed to q_{ik} , and the other features $h \in \mathcal{F} \setminus S$ stays x_{ih} . Hence the set of $\mathbf{z}_{iS}$ ($S \subseteq \mathcal{F}$) composes the domain of all possible values of $\mathbf{z}$. Define a corresponding scalar variable for any $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the value of the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ can be represented by a binary variable a_{iS} together with the scalar g_{iS} , namely,

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimization problem equation (13) is hence reformulated as a mixed integer programming below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \tag{14a}$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \tag{14b}$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \tag{14c}$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \tag{14d}$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \tag{14e}$$

The two constraints equation (14b) and equation (14c) make minimizing η equivalent to minimizing the absolute MeanD objective in equation (13). The constraints in equation (14d) assign each data point i to exactly one feature-modification plan a_{iS} for some $S \subseteq \mathcal{F}$, and equation (14e) enforces the same action budget as equation (13b). Solving equation (14) therefore gives the exact reference curve used in Figure 4 of Section 6.

H EXTENDED NUMERICAL RESULTS

The extended curves in Figures 5–8, with the individual panels in Figure 5 and Figure 8, confirm the main-text comparison of Shapley choices in lines 3–4. RB- p_{Uni} and CF- p_{Uni} differ only in A_{Shap} , and CF- p_{Uni} generally attains lower divergence for the same action budget. Similarly, RB- p_{OT} and CF- p_{OT} differ only in A_{Shap} , and CF- p_{OT} consistently dominates RB- p_{OT} . These matched comparisons show that RB-SHAP is not well aligned with the sparse CE-refinement objective.

Line 5 provides the second key comparison: how the joint probability $p(\mathbf{x}, \mathbf{r})$ defines the replacement matrix $\mathbf{q}$ and hence the direction of each selected edit, with $\mathbf{x}$ and $\mathbf{r}$ compared through the coupling $\mathbf{p}$. Across Figures 5–8, CF- p_{OT} is consistently the strongest method. The contrast with CF- p_{Uni} and CF- p_{Rnd} is especially informative because all three methods use counterfactual information but differ in the joint distribution $\mathbf{p}$ when exact alignment is unavailable. The resulting performance gaps show that counterfactual marginals alone are insufficient; the coupling structure materially affects the achieved effect for a fixed budget.

The evidence therefore favors OT when factual-counterfactual alignment is missing or potentially inaccurate. Even when a CE algorithm returns an explicit factual-counterfactual pairing, Figure 4 in Section 6 shows that this known pairing need not be the best alignment for sparse action selection.

The OT coupling p_{OT} should not be interpreted as the true data-generating joint distribution of $\mathbf{x}$ and $\mathbf{r}$. Its role is operational: it supplies a cost-aware alignment for both FA and CE refinement, so the empirical comparison evaluates how well edits selected from $\mathbf{x}$ toward $\mathbf{r}$ reduce the target divergence. The QDA column in Figure 5 further illustrates the stability of OT-based methods, while several non-OT baselines diverge from the target. At the same time, COLA is *not limited to using OT* as A_{Prob} ; because Figure 4 still shows a gap between available couplings and the global optimum, designing stronger coupling algorithms remains an important direction.

I EXPERIMENTS REPRODUCIBILITY

The experiments were run on a high-performance computing (HPC) cluster using four nodes in parallel, one for each dataset. Each node had two Intel Xeon Processor 2660v3 CPUs (10 cores, 2.60GHz) and 128 GB memory. Runtime was approximately 5–10 hours per node depending on dataset size; the same experiments can be reproduced on a laptop, but with longer wall-clock time.

For all four datasets, numerical features are standardized and categorical features are encoded with either label encoding or one-hot encoding. We did not observe a meaningful performance difference between these encoding choices for COLA. All experiments use a 7 : 3 train-test split.

The optimality baseline in Figure 4 was solved with Gurobi 11.0.2 ([gur](#)); reproducing that exact baseline requires a Gurobi license, although open-source operations-research libraries such as Google-OR tools ([goo](#)) can be used as alternatives. Because the MILP in Appendix G is computationally expensive, the exact benchmark is practical only for small-scale settings such as German Credit unless the feature set is reduced.

Model configurations are fixed across the comparisons. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA use scikit-learn implementations with default hyperparameters ([skl](#)). DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN has three layers and the SVM uses a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are used through their scikit-learn interfaces with default hyperparameters.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) only as general-purpose assist tools. Their role was limited to grammar, wording, and light copy-editing of author-written text. LLMs did not contribute research ideation, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Any LLM-suggested phrasing was reviewed and edited before inclusion in the paper. The authors take full responsibility for all content; LLMs are not authors or contributors. This disclosure complies with the ICLR policy on LLM usage.

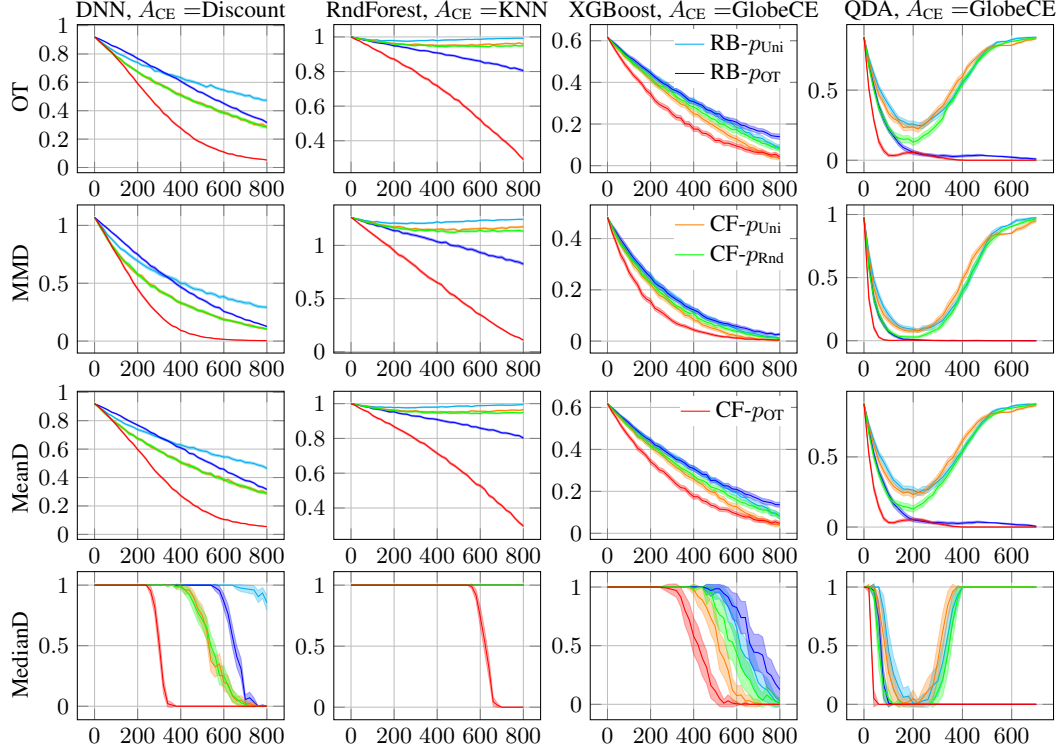


Figure 5: [HELOC] Remaining target divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as the action budget C increases. Each curve averages 100 runs and shadows show 99.9% confidence intervals; lower curves indicate stronger counterfactual recovery for the same number of edits. Legends apply to all plots, and $A_{\text{Value}}^{\text{avg}}$ is used.

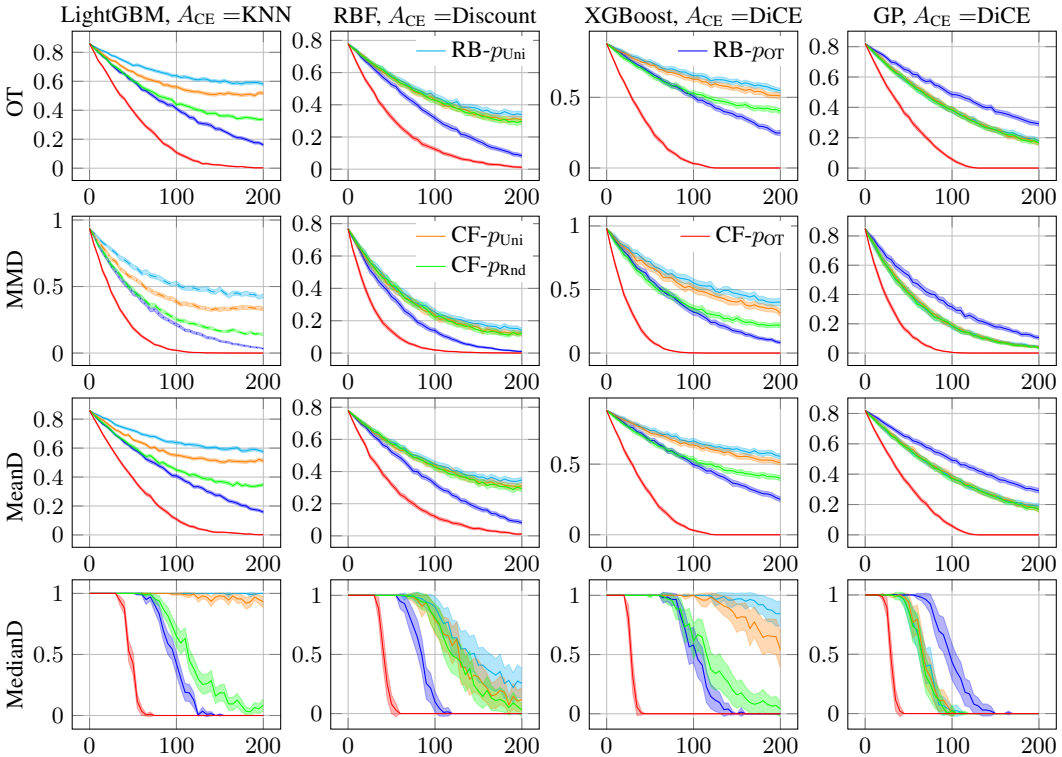


Figure 6: [German Credit] Remaining target divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as the action budget C increases. Each curve averages 100 runs and shadows show 99.9% confidence intervals; lower curves indicate stronger counterfactual recovery for the same number of edits. Legends apply to all plots, and $A_{\text{Value}}^{\text{avg}}$ is used.

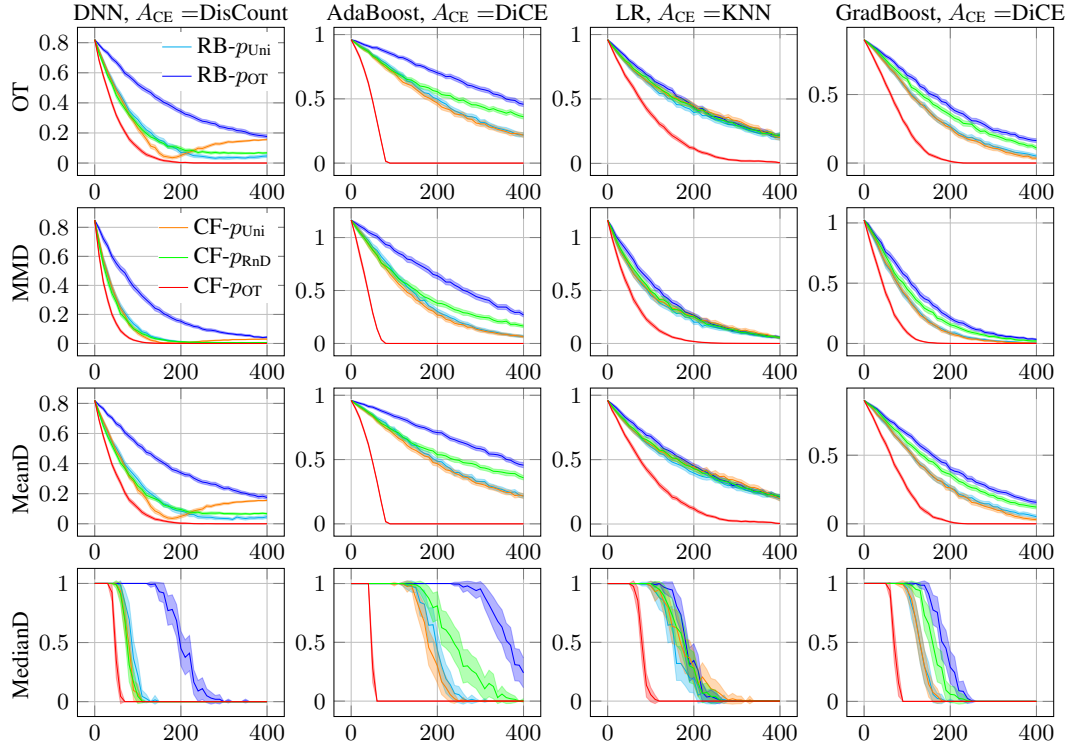


Figure 7: [Hotel Bookings] Remaining target divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as the action budget C increases. Each curve averages 100 runs and shadows show 99.9% confidence intervals; lower curves indicate stronger counterfactual recovery for the same number of edits. Legends apply to all plots, and $A_{\text{Value}}^{\max}$ is used.

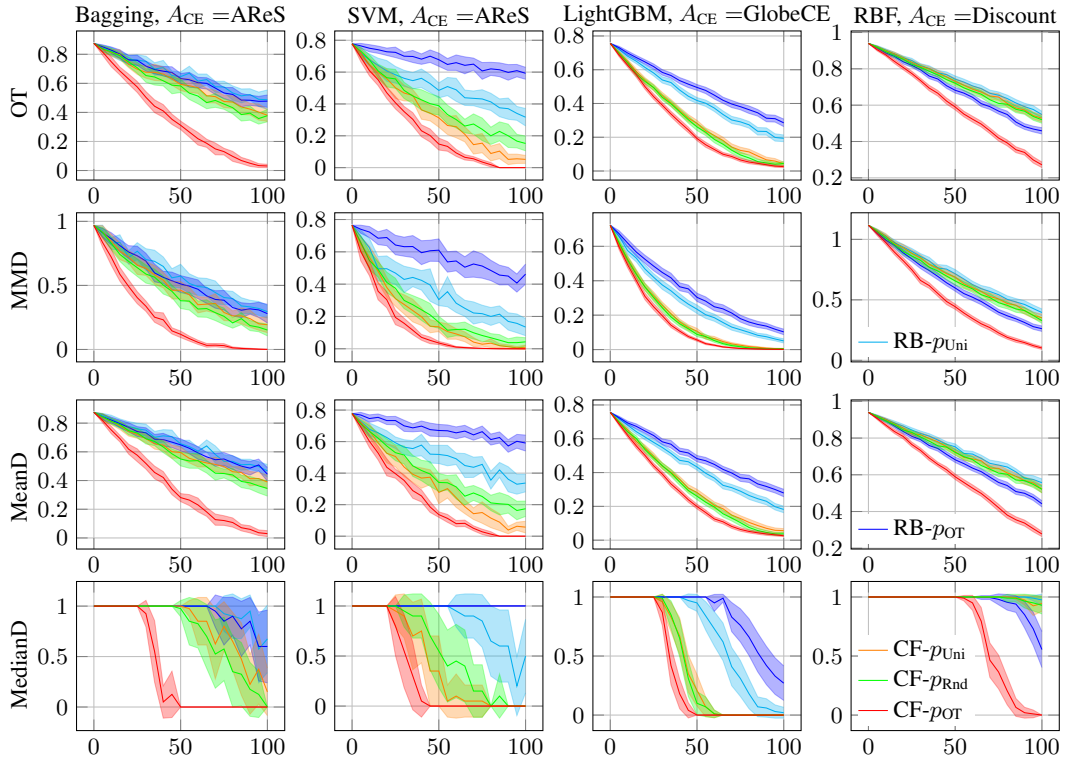


Figure 8: [COMPAS] Remaining target divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as the action budget C increases. Each curve averages 100 runs and shadows show 99.9% confidence intervals; lower curves indicate stronger counterfactual recovery for the same number of edits. Legends apply to all plots, and $A_{\text{Value}}^{\max}$ is used.